

Rest Energy from Density-Weighted Loop Curvature: A Covariant Locking Principle

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Abstract

We collect the loop-mass identities connecting rest energy E_{rest} and an effective wavenumber k_{eff} , prove the density-weighted locking principle that yields k_{eff} , and record extensions (holonomy, curvature, thickness, medium, dissipation, topology). We add stability, uniqueness, and quantitative error bounds for imperfect lock, and state the curved-space split explicitly.

1 Kinematics, Loop Geometry, and Weight

Conventions. Metric signature $(-, +, +, +)$ with Levi-Civita $\varepsilon_{0123} = +1$. Fix a unit timelike field u^μ ($u^\mu u_\mu = -1$) and orient the u -orthogonal 3-space so that the Frenet-Serret triad (t^μ, n^μ, b^μ) is right-handed: $\varepsilon_{\alpha\beta\gamma\delta} u^\alpha t^\beta n^\gamma b^\delta > 0$. Arc-length $s \in [0, L_s)$ is periodic, $s \sim s + L_s$.

Definition 1.1 (Background and induced geometry). Let $h_{\mu\nu} := g_{\mu\nu} + u_\mu u_\nu$ be the projector to the u -orthogonal 3-space. A closed spacelike loop $\Gamma \subset \{u \cdot x = \text{const}\}$ has arclength element $ds = \sqrt{h_{\mu\nu} dx^\mu dx^\nu}$ and total length $L_s := \oint_\Gamma ds$.

Definition 1.2 (Frenet-Serret data and geometric scalar). Define the unit tangent $t^\mu := dx^\mu/ds$ ($u_\mu t^\mu = 0$). In the induced 3-geometry, the Frenet-Serret frame (t^μ, n^μ, b^μ) obeys

$$\nabla_s t^\mu = \kappa n^\mu, \quad \nabla_s n^\mu = -\kappa t^\mu + \tau b^\mu, \quad \nabla_s b^\mu = -\tau n^\mu,$$

with $\nabla_s := t^\alpha h^\beta_\alpha \nabla_\beta$. Set the geometric scalar

$$\sigma(s) := \sqrt{\kappa(s)^2 + \tau(s)^2}.$$

Definition 1.3 (Phase, winding, and longitudinal gradient). Let $\phi(s)$ be the confined phase along Γ and $k_\parallel(s) := \partial_s \phi$ its longitudinal gradient. The winding number $n \in \mathbb{Z}$ is fixed by

$$\oint_\Gamma \partial_s \phi ds = 2\pi n,$$

with the sign determined by the orientation above.

Definition 1.4 (Linear energy density and measurement). The linear energy density along Γ is

$$w(s) := \int_{\Sigma_\perp(s)} T_{\mu\nu} u^\mu u^\nu d\Sigma,$$

where $\Sigma_\perp(s)$ is any small 2-surface orthogonal to both u^μ and t^μ . In experiments, w is obtained from sampled power/fluence on a calibrated cross-section; in simulations, from the stress-energy density integrated over a fixed-area slice.

Definition 1.5 (Weighted average). For any scalar $f(s)$ on Γ , the density-weighted average is

$$\langle f \rangle_w := \frac{\oint_{\Gamma} w(s) f \, ds}{\oint_{\Gamma} w(s) \, ds} := \frac{\oint_{\Gamma} w(s) f(s) \, ds}{\oint_{\Gamma} w(s) \, ds}.$$

Standing assumptions. Γ is C^2 and closed; $\sigma \in L^2(\Gamma)$; $w \in L^\infty(\Gamma)$ with $w(s) \geq w_{\min} > 0$ a.e. Variations η of the phase preserve winding:

$$\oint_{\Gamma} \partial_s \eta \, ds = 0.$$

All statements are invariant under C^1 reparameterizations of s .

Sampling model (for data/estimation). Let $\{s_i\}_{i=1}^N$ be uniform samples with $\Delta s = L_s/N$. Measured arrays $\{\kappa_i, \tau_i, w_i\}$ approximate $\{\kappa(s_i), \tau(s_i), w(s_i)\}$ with bounded noise. Define $\sigma_i := \sqrt{\kappa_i^2 + \tau_i^2}$ and the consistent Riemann estimators

$$\widehat{L}_s = \sum_i \Delta s, \quad \widehat{\langle f \rangle}_w = \frac{\sum_i w_i f_i \Delta s}{\sum_i w_i \Delta s},$$

which converge to the continuum definitions as $\Delta s \rightarrow 0$.

2 Energy Identity and Canonical Forms

Convention. $k_{\text{eff}} \geq 0$ is the (scalar) effective wavenumber associated to the closed loop Γ . The winding $n \in \mathbb{Z}$ is signed by the chosen orientation; whenever it appears in k_{eff} we use $|n|$ to keep $E_{\text{rest}} \geq 0$.

Definition 2.1 (Rest-energy identity).

$$E_{\text{rest}} = mc^2 = \hbar c k_{\text{eff}}. \quad (2.1)$$

F1: Topological-length (exact, orientation-agnostic).

$$k_{\text{eff}} = \frac{2\pi|n|}{L_s}, \quad \Rightarrow \quad m = \frac{\hbar}{c} \frac{2\pi|n|}{L_s}. \quad (2.2)$$

F2: Geometric average (unweighted).

$$k_{\text{eff}} = \frac{1}{L_s} \oint_{\Gamma} \sigma \, ds = \langle \sigma \rangle_{\Gamma}, \quad \Rightarrow \quad m = \frac{\hbar}{c} \langle \sigma \rangle_{\Gamma}. \quad (2.3)$$

Planar reduction. If $\tau \equiv 0$ and Γ is a regular plane loop with turning index $w \in \mathbb{Z}$, then $\oint_{\Gamma} \kappa \, ds = 2\pi w$. For a simple convex loop ($w = 1$) we have $k_{\text{eff}} = \frac{2\pi}{L_s}$, i.e. F2 $\rightarrow$ F1.

F3: Covariant, density-weighted (measurement form).

$$k_{\text{eff}} = \langle \sigma \rangle_w = \frac{\oint_{\Gamma} w(s) \sigma \, ds}{\oint_{\Gamma} w(s) \, ds} = \frac{\oint_{\Gamma} w(s) \sigma(s) \, ds}{\oint_{\Gamma} w(s) \, ds}. \quad (2.4)$$

F4: Phase–gradient (weighted).

$$k_{\text{eff}} = \langle k_{\parallel} \rangle_w = \frac{\oint_{\Gamma} w(s) \partial_s \phi \, ds}{\oint_{\Gamma} w(s) \, ds} = \frac{\oint_{\Gamma} w(s) \partial_s \phi(s) \, ds}{\oint_{\Gamma} w(s) \, ds}. \quad (2.5)$$

Equivalence and bounds (used later).

Lemma 2.2 (Locking $\Rightarrow$ F3=F4). *If the variational lock holds, $k_{\parallel}(s) = \sigma(s)$ a.e. on Γ , then $\langle \sigma \rangle_w = \langle k_{\parallel} \rangle_w$, hence (2.4) and (2.5) coincide.*

Proposition 1 (Universal lower bounds). *For any C^2 closed loop and any admissible w with $w_{\min} \leq w \leq w_{\max}$,*

$$\langle \sigma \rangle_{\Gamma} \geq \frac{2\pi}{L_s} \quad (\text{Fenchel}), \quad (2.6)$$

and

$$\langle \sigma \rangle_w \geq \frac{w_{\min}}{\langle w \rangle_{\Gamma}} \langle \sigma \rangle_{\Gamma} \geq \frac{w_{\min}}{\langle w \rangle_{\Gamma}} \frac{2\pi}{L_s}, \quad (2.7)$$

where $\langle w \rangle_{\Gamma} := L_s^{-1} \oint_{\Gamma} w \, ds$.

Notes. (i) F1 is purely topological/metric (length + winding); (ii) F2 injects geometry via σ and reduces to F1 in planar cases; (iii) F3 is the experimentally natural estimator; (iv) F4 is the phase–space form, equal to F3 under locking (Lemma 2.2). The sign of n enters holonomy shifts (treated later); the rest energy uses $|n|$.

3 Variational Locking and Consequences

Convention. Use the geometric scalar $\sigma(s) := \sqrt{\kappa(s)^2 + \tau(s)^2}$ and weight $w(s) > 0$ a.e. on Γ . The winding $n \in \mathbb{Z}$ is fixed by orientation.

Definition 3.1 (Effective line action).

$$\mathcal{S}_{\text{eff}}[\phi; \Gamma] = \oint_{\Gamma} w(s) (\partial_s \phi - \sigma(s))^2 \, ds + \Lambda \left(\oint_{\Gamma} \partial_s \phi \, ds - 2\pi n \right). \quad (3.1)$$

Proposition 2 (Stationarity; exact Euler–Lagrange form). *Critical points of (3.1) satisfy*

$$\partial_s (2w(\partial_s \phi - \sigma) + \Lambda) = 0 \quad \Longleftrightarrow \quad \partial_s \phi = \sigma + \frac{\alpha}{w}, \quad (3.2)$$

for a constant $\alpha \in \mathbb{R}$. The winding constraint yields

$$\alpha \oint_{\Gamma} \frac{ds}{w(s)} = 2\pi n - \oint_{\Gamma} \sigma \, ds \quad \Longrightarrow \quad \alpha = \frac{2\pi n - \oint_{\Gamma} \sigma \, ds}{\oint_{\Gamma} \frac{ds}{w(s)}}. \quad (3.3)$$

Proposition 3 (Global minimum, perfect lock iff closure holds). *Among all C^1 phases obeying the winding constraint, the unique (mod 2π) minimizer of (3.1) is (3.2) with α in (10.4). The minimum value is*

$$\min \mathcal{S}_{\text{eff}} = \frac{(2\pi n - \oint_{\Gamma} \sigma \, ds)^2}{\oint_{\Gamma} \frac{ds}{w(s)}}. \quad (3.4)$$

In particular, perfect lock $\partial_s \phi = \sigma$ occurs iff the closure condition $\oint_{\Gamma} \sigma \, ds = 2\pi n$ holds; then $\alpha = 0$ and $\min \mathcal{S}_{\text{eff}} = 0$.

Corollary 1 (Recovery of F3 under lock). *If the lock holds, $\partial_s \phi = \sigma$ and hence $k_{\text{eff}} = \langle \sigma \rangle_w$ (F3). Combined with (2.1), $E_{\text{rest}} = \hbar c \langle \sigma \rangle_w$.*

Corollary 2 (Constant-weight reduction $w \equiv w_0$). *If w is constant, then the minimizer has $\partial_s \phi - \sigma = \alpha/w_0$ (a constant) and*

$$\langle \partial_s \phi \rangle_w = \frac{2\pi n}{L_s} \implies k_{\text{eff}} = \langle k_{\parallel} \rangle_w = \frac{2\pi n}{L_s} \quad (\text{F4 reduces to F1}). \quad (3.5)$$

Equality $\text{F4} = \text{F3}$ occurs iff $\oint_{\Gamma} \sigma \, ds = 2\pi n$.

Theorem 3.2 (Strict stability and uniqueness). *For $\phi \rightarrow \phi + \varepsilon \eta$ with $\oint_{\Gamma} \partial_s \eta \, ds = 0$,*

$$\delta^2 \mathcal{S}_{\text{eff}} = 2 \oint_{\Gamma} w(s) (\partial_s \eta)^2 \, ds > 0 \quad (3.6)$$

unless η is constant. Thus the minimizer in Prop. 3 is unique modulo a global phase.

Corollary 3 (Quantitative mislock error). *Let $\delta(s) := \partial_s \phi - \sigma(s) = \alpha/w(s)$ as in (3.2)–(10.4). Then*

$$|k_{\text{eff}} - \langle \sigma \rangle_w| = \langle \delta \rangle_w = \frac{\alpha L_s}{\oint_{\Gamma} w \, ds} = \frac{L_s}{\oint_{\Gamma} w \, ds} \frac{2\pi n - \oint_{\Gamma} \sigma \, ds}{\oint_{\Gamma} \frac{ds}{w(s)}}, \quad (3.7)$$

and, by Cauchy–Schwarz, $|\langle \delta \rangle_w| \leq \sqrt{\frac{\oint_{\Gamma} w \delta^2 \, ds}{\oint_{\Gamma} w \, ds}}$. Consequently $|m - m_0| \leq \frac{\hbar}{c} \sqrt{\frac{\oint_{\Gamma} w \delta^2 \, ds}{\oint_{\Gamma} w \, ds}}$, with $m_0 := \frac{\hbar}{c} \langle \sigma \rangle_w$.

Definition 3.3 (Holonomy fields and labels). Let q be the charge, A_{μ} a $U(1)$ potential with flux $\Phi := \oint_{\Gamma} A_{\mu} t^{\mu} \, ds$ and flux quantum $\Phi_0 := h/q$. Let Ω_{μ} be the spin connection pulled back to Γ , and set the dimensionless spin holonomy $\Gamma_s := \frac{1}{2\pi} \oint_{\Gamma} \Omega_{\mu} t^{\mu} \, ds$. The spin label is $s \in \frac{1}{2}\mathbb{Z}$.

Proposition 4 (Gauge/spin-covariant extension). *Replace $\partial_s \phi$ by the covariant derivative*

$$D_s \phi := \partial_s \phi - \frac{q}{\hbar} A_{\mu} t^{\mu} - s \Omega_{\mu} t^{\mu}. \quad (3.8)$$

Then all results above hold with $\partial_s \phi$ replaced by $D_s \phi$, the winding constraint becomes

$$\oint_{\Gamma} D_s \phi \, ds = 2\pi n \iff \oint_{\Gamma} \partial_s \phi \, ds = 2\pi n + \frac{q}{\hbar} \Phi + 2\pi s \Gamma_s, \quad (3.9)$$

and the lock condition reads $D_s \phi = \sigma$. For $w \equiv w_0$, the averaged phase gradient satisfies $\langle \partial_s \phi \rangle_w = \frac{2\pi n}{L_s} + \frac{q}{\hbar} \frac{\Phi}{L_s} + s \frac{2\pi \Gamma_s}{L_s}$ (holonomy-shifted F1).

Gauge/Spin-Covariant Locking

Conventions. $\sigma(s) := \sqrt{\kappa^2 + \tau^2}$, $w(s) > 0$ a.e. on Γ . Charge q , $U(1)$ potential A_{μ} , spin connection Ω_{μ} pulled back to Γ , spin label $s \in \frac{1}{2}\mathbb{Z}$. Define the covariant phase gradient

$$D_s \phi := \partial_s \phi - \frac{q}{\hbar} A_{\mu} t^{\mu} - s \Omega_{\mu} t^{\mu}, \quad (3.10)$$

and holonomies $\Phi := \oint_{\Gamma} A_{\mu} t^{\mu} \, ds$, $\Phi_0 := h/q$, $\Gamma_s := \frac{1}{2\pi} \oint_{\Gamma} \Omega_{\mu} t^{\mu} \, ds$.

Definition 3.4 (Gauge/Spin-covariant line action).

$$\mathcal{S}_{\text{eff}}^{\text{cov}}[\phi; \Gamma] = \oint_{\Gamma} w(s) (D_s \phi - \sigma(s))^2 ds + \Lambda \left(\oint_{\Gamma} D_s \phi ds - 2\pi n \right). \quad (3.11)$$

Proposition 5 (Covariant Euler–Lagrange and minimizer). *Stationarity of (3.11) yields*

$$\partial_s(2w(D_s \phi - \sigma) + \Lambda) = 0 \implies D_s \phi = \sigma + \frac{\alpha}{w}, \quad (3.12)$$

with

$$\alpha \oint_{\Gamma} \frac{ds}{w(s)} = 2\pi n - \oint_{\Gamma} \sigma ds, \quad \min \mathcal{S}_{\text{eff}}^{\text{cov}} = \frac{(2\pi n - \oint_{\Gamma} \sigma ds)^2}{\oint_{\Gamma} \frac{ds}{w(s)}}. \quad (3.13)$$

Thus perfect lock $D_s \phi = \sigma$ occurs iff $\oint_{\Gamma} \sigma ds = 2\pi n$.

Corollary 4 (Recovery of F3 under lock; averaged phase). *Under perfect lock, $D_s \phi = \sigma$ and*

$$\langle \partial_s \phi \rangle_w = \langle \sigma \rangle_w + \frac{q}{\hbar} \langle A_{\mu} t^{\mu} \rangle_w + s \langle \Omega_{\mu} t^{\mu} \rangle_w. \quad (3.14)$$

Hence $k_{\text{eff}} = \langle k_{\parallel} \rangle_w = \langle \partial_s \phi \rangle_w$ reproduces the holonomy shifts. For uniform weight $w \equiv w_0$,

$$k_{\text{eff}} = \frac{2\pi}{L_s} \left(n + \frac{\Phi}{\Phi_0} + s \Gamma_s \right) \quad (\text{holonomy-shifted F1}). \quad (3.15)$$

Corollary 5 (Phase budget / quantization). *The winding constraint in covariant form, $\oint_{\Gamma} D_s \phi ds = 2\pi n$, is equivalent to*

$$\oint_{\Gamma} \partial_s \phi ds = 2\pi n + \frac{q}{\hbar} \Phi + 2\pi s \Gamma_s. \quad (3.16)$$

Under perfect lock, integrating $D_s \phi = \sigma$ gives $\oint_{\Gamma} \sigma ds = 2\pi n - \frac{q}{\hbar} \Phi - 2\pi s \Gamma_s + \oint_{\Gamma} (\partial_s \phi - \sigma) ds = 2\pi n$, consistent with (3.13).

Invariance check. Under $U(1)$ gauge ($A_{\mu} t^{\mu} \mapsto A_{\mu} t^{\mu} + \partial_s \chi$, $\phi \mapsto \phi + \frac{q}{\hbar} \chi$) and spin-frame rotation ($\Omega_{\mu} t^{\mu} \mapsto \Omega_{\mu} t^{\mu} + \partial_s \theta$, $\phi \mapsto \phi + s\theta$), $D_s \phi$, the constraint, and $\mathcal{S}_{\text{eff}}^{\text{cov}}$ are invariant.

Mislock with holonomy. If $\oint_{\Gamma} \sigma ds \neq 2\pi n$, then $\delta := D_s \phi - \sigma = \alpha/w$ with α from (3.13); the error bounds of Cor. 3 apply verbatim.

4 Dualities, Estimators, and Sensitivities

F6: Compton/Length Duality.

$$L_s = n \lambda_C, \quad \lambda_C = \frac{h}{mc} \iff m = \frac{2\pi n \hbar}{c L_s}. \quad (4.1)$$

F7: Discrete Estimator (Polygonal Sampling). With samples $(s_i, \Delta s_i)$, curvatures κ_i , torsions τ_i , and weights w_i ,

$$\widehat{k_{\text{eff}}} = \frac{\sum_i w_i \sqrt{\kappa_i^2 + \tau_i^2} \Delta s_i}{\sum_i w_i \Delta s_i}, \quad \widehat{m} = \frac{\hbar}{c} \widehat{k_{\text{eff}}}. \quad (4.2)$$

F8: Tension/Shape Sensitivity (from F1).

$$E(L_s) = \hbar c \frac{2\pi n}{L_s}, \quad \frac{dE}{dL_s} = -\hbar c \frac{2\pi n}{L_s^2} =: -\mathcal{T}. \quad (4.3)$$

F9: Composite Energy (Non-Interacting Loops). $E_{\text{rest}}^{\text{tot}} = \sum_a \hbar c k_{\text{eff}}^{(a)}$, hence $m_{\text{tot}} = \frac{1}{c^2} \sum_a \hbar c k_{\text{eff}}^{(a)}$ (plus interactions/strain if present).

5 Extensions (Holonomy, Curvature, Thickness, Medium, Dissipation, Topology)

F10/F19: Exact Holonomy Shift (Gauge/Spin). Aharonov–Bohm phase $\Delta\phi_{\text{AB}} = \frac{q}{\hbar} \oint A_\mu t^\mu ds = 2\pi \Phi/\Phi_0$ with $\Phi_0 = h/q$; spin-connection phase $\oint \Omega_\mu t^\mu ds = 2\pi \Gamma_s$. Then

$$k_{\text{eff}} = \frac{2\pi}{L_s} \left(n + \frac{\Phi}{\Phi_0} + \mathfrak{s} \Gamma_s \right) \quad (\text{exact shift}). \quad (5.1)$$

With density weighting and geometry,

$$k_{\text{eff}} = \left\langle \sigma + \frac{q}{\hbar} A_\mu t^\mu + \mathfrak{s} \Omega_\mu t^\mu \right\rangle_w.$$

F11/F20: Curved Spacetime Split. In the u -orthogonal 3-geometry with Riemann tensor ${}^{(3)}R_{ijkl}$,

$$\sigma^2 = \kappa_g^2 + \tau^2 + \mathcal{K}_{\text{ext}}, \quad \mathcal{K}_{\text{ext}} := n^i n^j {}^{(3)}R_{ikjl} t^k t^l, \quad (5.2)$$

so $k_{\text{eff}} = \left\langle \sqrt{\kappa_g^2 + \tau^2 + \mathcal{K}_{\text{ext}}} \right\rangle_w$.

Remark 1 (Geodesic vs extrinsic curvature). Here κ_g is the geodesic curvature of Γ within the u -orthogonal 3-geometry, while $\mathcal{K}_{\text{ext}} := n^i n^j {}^{(3)}R_{ikjl} t^k t^l$ encodes ambient 3-curvature along (t^i, n^i) .

F12: Worksheet Reformulation. With worldsheet metric γ_{ab} and lineal tension density $\mathcal{T}(s) := \hbar c w(s) \sigma(s) / \oint w ds$,

$$S_{\text{ws}} = \int_{\mathcal{W}} \sqrt{-\gamma} \mathcal{T}(s) d^2 \xi, \quad (5.3)$$

whose on-shell energy equals $\hbar c k_{\text{eff}}$. Variation in $g_{\mu\nu}$ yields a localized $T^{\mu\nu}$.

F13: Ribbon Correction (Twist–Writhe) [conj]. For a framed loop with $Lk = Tw + Wr \in \mathbb{Z}$,

$$k_{\text{eff}} = \frac{2\pi}{L_s} (n + Lk) \quad (\text{reduces to F1 for } Lk = 0). \quad (5.4)$$

F14: Finite-Thickness (Thin-Tube Expansion). For tube radius $a \ll R_{\text{curv}}$,

$$k_{\text{eff}}(a) = \left\langle \sigma(1 + c_2 a^2 \kappa^2 + c'_2 a^2 \tau^2 + O(a^4)) \right\rangle_w, \quad c_2, c'_2 = O(1). \quad (5.5)$$

F15: Non-Null Dispersion (Medium/Interaction). If the local phase speed satisfies $v_{\text{ph}}(s) \leq c$,

$$E = \hbar \left\langle v_{\text{ph}} k_{\parallel} \right\rangle_w, \quad m = \frac{E}{c^2} = \frac{\hbar}{c^2} \left\langle v_{\text{ph}} k_{\parallel} \right\rangle_w. \quad (5.6)$$

F16: Radiation/Q-Factor Drift. With $\omega = c k_{\text{eff}}$ and quality factor Q ,

$$P_{\text{loss}} = \frac{E \omega}{Q}, \quad \dot{m} = -\frac{P_{\text{loss}}}{c^2} = -\frac{\hbar k_{\text{eff}} \omega}{Q c}. \quad (5.7)$$

F18: Topological Lower Bounds (Knots). Fenchel/Fáry–Milnor give $\oint_{\Gamma} \kappa \, ds \geq 2\pi$ (all loops), and $\geq 4\pi$ (nontrivial knots). For planar $\tau = 0$,

$$m \geq \frac{\hbar}{c} \frac{2\pi}{L_s} \quad (\text{unknotted}), \quad m \geq \frac{\hbar}{c} \frac{4\pi}{L_s} \quad (\text{knotted}). \quad (5.8)$$

6 Relativistic and Quantum Consistency

Definitions. Four-wavevector $K^\mu := (\omega/c, \mathbf{k})$, four-momentum $P^\mu := (E/c, \mathbf{p})$ with $E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$. Let $k := \|\mathbf{k}\|$, $\lambda := 2\pi/k$, and use k_{eff} from (2.1) as the rest scale.

F21: de Broglie relations.

$$E = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k}, \quad \lambda = \frac{2\pi}{k} = \frac{h}{p}. \quad (6.1)$$

F22: Massive dispersion (Klein–Gordon form with k_{eff}).

$$\omega^2 = c^2(k^2 + k_{\text{eff}}^2), \quad k_{\text{eff}} = \frac{mc}{\hbar}. \quad (6.2)$$

Theorem 6.1 (Equivalence to Einstein). *With eqs. (6.1) and (6.2) and (2.1), one has*

$$E^2 = (pc)^2 + (mc^2)^2. \quad (6.3)$$

Proof. Multiply eq. (6.2) by $\hbar^2$: $(\hbar\omega)^2 = c^2((\hbar k)^2 + (\hbar k_{\text{eff}})^2)$. Using $E = \hbar\omega$, $p = \hbar k$, $mc = \hbar k_{\text{eff}}$ gives eq. (6.3). $\square$

F23: Lorentz invariant (covariant statement).

$$K_\mu K^\mu = \left(\frac{\omega}{c}\right)^2 - k^2 = k_{\text{eff}}^2 \iff P_\mu P^\mu = \left(\frac{E}{c}\right)^2 - p^2 = (mc)^2. \quad (6.4)$$

Proposition 6 (Group velocity equals particle velocity). *From eq. (6.2), $v_g = \partial\omega/\partial k = \frac{c^2 k}{\omega} = \frac{pc^2}{E}$.*

Corollary 6 (Nonrelativistic limit). *For $k \ll k_{\text{eff}}$,*

$$\hbar\omega = mc^2 + \frac{p^2}{2m} + O\left(\frac{p^4}{m^3 c^2}\right). \quad (6.5)$$

F24: Compton scales (rest frame).

$$\omega_C = c k_{\text{eff}} = \frac{mc^2}{\hbar}, \quad \lambda_C = \frac{2\pi}{k_{\text{eff}}} = \frac{h}{mc}. \quad (6.6)$$

Consistency with locking. If the variational lock (Prop. ??) gives $k_{\text{eff}} = \langle \sigma \rangle_w$, then eq. (6.2) reads

$$\omega^2 = c^2 \left(k^2 + (\langle \sigma \rangle_w)^2 \right), \quad (6.7)$$

which, via theorem 6.1, returns eq. (6.3) with $m = (\hbar/c) \langle \sigma \rangle_w$.

7 Units, Limits, and Worked Examples

Units. $\kappa, \tau, \sigma, k_{\parallel}, k_{\text{eff}} : \text{m}^{-1}$; $s, L_s : \text{m}$; $w : \text{J m}^{-1}$ (linear energy density); $m : \text{kg}$; $E_{\text{rest}} : \text{J}$; $\hbar : \text{J s}$; $c : \text{m s}^{-1}$; $\Phi, \Phi_0 : \text{Wb}$; $\Gamma_s : \text{dimensionless}$.

Symbol	Units	Meaning
s, L_s	m	arclength, loop length
κ, τ	m^{-1}	curvature, torsion
$\sigma = \sqrt{\kappa^2 + \tau^2}$	m^{-1}	geometric scalar
$k_{\parallel} = \partial_s \phi$	m^{-1}	longitudinal phase gradient
$k_{\text{eff}} = \langle k_{\parallel} \rangle_w$	m^{-1}	effective wavenumber
$w(s)$	J m^{-1}	linear energy density (weight)
m, E_{rest}	kg, J	rest mass, rest energy
$\hbar, c$	J s, m s^{-1}	constants
Φ, Φ_0, Γ_s	Wb, Wb, 1	flux, flux quantum, spin holonomy

Limits. (i) $L_s \rightarrow \infty \Rightarrow k_{\text{eff}} \rightarrow 0$ and $m \rightarrow 0$. (ii) Uniform $w \Rightarrow \text{F3} \rightarrow \text{F2}$; planar convex loops ($\tau = 0$) give $\text{F2} \rightarrow \text{F1}$ for $n = 1$. (iii) Fenchel/Fáry–Milnor floors: $\oint \sigma ds \geq 2\pi$ (unknotted), $\geq 4\pi$ (knotted).

Example A: Planar Circle (Uniform w)

Let $R = 0.5 \text{ m}$. Then $\kappa = 1/R$, $\tau = 0$, $\sigma = \kappa$, so $k_{\text{eff}} = 1/R$ and

$$m = \frac{\hbar}{cR} \approx \frac{1.054 \times 10^{-34}}{(2.998 \times 10^8)(0.5)} = 7.03 \times 10^{-43} \text{ kg}, \quad E_{\text{rest}} = mc^2 \approx 6.32 \times 10^{-26} \text{ J}.$$

Example B: Helix (Uniform w)

For radius r and pitch per turn p , one turn has $L_s = \sqrt{(2\pi r)^2 + p^2}$ and

$$\kappa = \frac{r}{r^2 + (p/2\pi)^2}, \quad \tau = \frac{(p/2\pi)}{r^2 + (p/2\pi)^2}, \quad \sigma = \frac{1}{\sqrt{r^2 + (p/2\pi)^2}}.$$

Hence $\oint \sigma ds = \sigma L_s = 2\pi$ ($n = 1$), so $m = \frac{\hbar}{c} \sigma$. For $r = 0.5 \text{ m}$, $p = 0.5 \text{ m}$:

$$\sigma \approx \frac{1}{0.5063} \text{ m}^{-1}, \quad m \approx 6.94 \times 10^{-43} \text{ kg}, \quad E_{\text{rest}} \approx 6.24 \times 10^{-26} \text{ J}.$$

Example C: Planar Ellipse with Nonuniform Weight ($\tau \equiv 0$)

Let $x = a \cos \theta$, $y = b \sin \theta$ with $a = 0.6$ m, $b = 0.4$ m. Then $ds = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} d\theta$ and

$$\kappa(\theta) = \frac{ab}{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}.$$

- Unweighted:

$$k_{\text{eff}}^{(0)} = \frac{1}{L_s} \oint \kappa ds = \frac{2\pi}{L_s}$$

with $L_s \approx 3.173$ m, so $k_{\text{eff}}^{(0)} \approx 1.980 \text{ m}^{-1}$ (F1/F2 agree).

- Nonuniform weight $w(\theta) = 1 + 0.2 \cos 4\theta$:

$$k_{\text{eff}}^{(w)} = \frac{\oint w \kappa ds}{\oint w ds} \approx 1.998 \text{ m}^{-1},$$

a +0.9% shift vs. unweighted. The induced mass/energy shifts are

$$\delta m = \frac{\hbar}{c} \delta k_{\text{eff}} \approx 6.26 \times 10^{-45} \text{ kg}, \quad \delta E = \hbar c \delta k_{\text{eff}} \approx 5.62 \times 10^{-28} \text{ J}.$$

8 Predictions and Experimental Levers

1. **Length scaling (planar, uniform):**

$$E_{\text{rest}} = \hbar c \frac{2\pi n}{L_s} \quad \Rightarrow \quad \frac{\Delta E}{E} = -\frac{\Delta L_s}{L_s}.$$

2. **Torsion penalty (fixed L_s & κ -profile):** increasing τ raises $\sigma = \sqrt{\kappa^2 + \tau^2}$ pointwise $\Rightarrow E_{\text{rest}}$ grows with $\langle \tau^2 \rangle$.

3. **Topological floors:**

$$E_{\text{rest}} \geq \hbar c \frac{2\pi}{L_s} \quad (\text{unknotted}), \quad E_{\text{rest}} \geq \hbar c \frac{4\pi}{L_s} \quad (\text{knotted}).$$

4. **State-dependent shifts (reweighting):**

$$\delta m = \frac{\hbar}{c} \delta \langle \sigma \rangle_w;$$

biasing w toward high- σ segments increases m .

5. **Holonomy (AB/spin) slopes (uniform w):**

$$k_{\text{eff}} = \frac{2\pi}{L_s} \left(n + \frac{\Phi}{\Phi_0} + s \Gamma_s \right), \quad \frac{dE}{d\Phi} = \hbar c \frac{2\pi}{L_s} \frac{1}{\Phi_0}.$$

9 Symbols and assumptions

Γ : smooth closed loop, $s \in [0, L)$, $L = \oint_{\Gamma} ds$,

$\hat{\mathbf{t}}(s) := \frac{d\mathbf{x}}{ds}$ (unit tangent), $\psi(s, t) = A(s, t) e^{i\phi(s, t)}$, $k_{\parallel} := \partial_s \phi$,

$k_{\text{eff}} := \frac{1}{L} \oint_{\Gamma} k_{\parallel} ds$, $w(s) > 0$, $\langle f \rangle_w := \frac{\oint_{\Gamma} w f ds}{\oint_{\Gamma} w ds}$,

$\kappa_g(s)$: geodesic curvature, $\tau(s)$: torsion,

$K_{\text{ext}}(s) := R_{ikjl}(x(s)) t^i n^k t^j n^l$ (ambient sectional curvature),

$\sigma(s) := \sqrt{\kappa_g(s)^2 + \tau(s)^2 (+K_{\text{ext}}(s))}$ (geometric rate),

$\mathbf{A}$: U(1) gauge potential, q : charge, $\Phi := \oint_{\Gamma} \mathbf{A} \cdot d\mathbf{l}$, $\Phi_0 := \frac{h}{q}$,

$S \in \frac{1}{2}\mathbb{Z}$ (spin index), ω_s : spin connection along Γ , $\Gamma_s := \frac{1}{2\pi} \oint_{\Gamma} \omega_s ds = \frac{1}{2\pi} \left(\int_D K dA + \oint_{\Gamma} \kappa_g ds \right)$,

c : speed of light (vacuum), $\hbar$: Planck constant.

Budget (covariant closure). With covariant tangential derivative $D_s \phi := \partial_s \phi - \frac{q}{\hbar} \mathbf{A} \cdot \hat{\mathbf{t}} - S \omega_s$ and single-valued phase,

$$\oint_{\Gamma} D_s \phi ds = 2\pi n \implies \oint_{\Gamma} \partial_s \phi ds = 2\pi \left(n + \frac{\Phi}{\Phi_0} + S \Gamma_s \right) =: B. \quad (9.1)$$

10 Lock variational principle

Define the lock functional

$$F[\phi; \Gamma] = \frac{\hbar c}{2} \oint_{\Gamma} w(s) (\partial_s \phi - \sigma)^2 ds. \quad (10.1)$$

Euler–Lagrange under the budget. Vary ϕ with periodic boundary conditions and enforce (9.1). Integration by parts gives the constant–mislock form

$$-\partial_s [w(\partial_s \phi - \sigma)] = 0 \implies w(\partial_s \phi - \sigma) = \alpha, \quad \alpha \in \mathbb{R} \text{ (constant)}. \quad (10.2)$$

Hence

$$\partial_s \phi = \sigma + \frac{\alpha}{w}, \quad \oint_{\Gamma} \partial_s \phi ds = \oint_{\Gamma} \sigma ds + \alpha \oint_{\Gamma} \frac{ds}{w}, \quad (10.3)$$

and α is fixed by the holonomy budget:

$$\alpha = \frac{B - \oint_{\Gamma} \sigma ds}{\oint_{\Gamma} \frac{ds}{w}}. \quad (10.4)$$

11 Two-rail equivalence and energy laws

Theorem 11.1 (Two rails and their equality). *Assume vacuum dispersion $\omega = c k_{\text{eff}}$. Then:*

$$(\textit{Topological/holonomy rail}) \quad k_{\text{eff}} = \frac{1}{L} \oint_{\Gamma} \partial_s \phi \, ds = \frac{2\pi}{L} \left(n + \frac{\Phi}{\Phi_0} + S \Gamma_s \right), \quad (11.1)$$

$$(\textit{Geometric/lock rail}) \quad k_{\text{eff}} = \left\langle \partial_s \phi \right\rangle_w = \left\langle \sigma \right\rangle_w + \alpha \frac{L}{\oint_{\Gamma} w \, ds}, \quad (11.2)$$

with α from (10.4). The following are equivalent:

$$\boxed{\oint_{\Gamma} \sigma \, ds = 2\pi \left(n + \frac{\Phi}{\Phi_0} + S \Gamma_s \right)} \iff \boxed{\alpha = 0} \iff \boxed{k_{\text{eff}} = \left\langle \sigma \right\rangle_w}. \quad (11.3)$$

Consequently, the energy/mass laws are

$$\boxed{E = \hbar \omega = \hbar c k_{\text{eff}}, \quad E L = 2\pi \hbar c \left(n + \frac{\Phi}{\Phi_0} + S \Gamma_s \right);} \quad \boxed{m = \frac{E}{c^2} = \frac{\hbar}{c} \left\langle \sigma \right\rangle_w \quad (\textit{at lock}).} \quad (11.4)$$

Mislock metrics (exact). Let $\delta := \partial_s \phi - \sigma$. From (10.2),

$$\delta = \frac{\alpha}{w}, \quad k_{\text{eff}} - \left\langle \sigma \right\rangle_w = \left\langle \delta \right\rangle_w = \alpha \frac{L}{\oint_{\Gamma} w \, ds},$$

and

$$\left\langle \delta^2 \right\rangle_w^{1/2} = \left| \alpha \right| \sqrt{\frac{\oint_{\Gamma} w^{-1} ds}{\oint_{\Gamma} w \, ds}}, \quad |m - m_{\text{lock}}| = \frac{\hbar}{c} \left\langle \delta^2 \right\rangle_w^{1/2}. \quad (11.5)$$

Flux quantum and slope.

$$\Phi_0 = \frac{h}{q}, \quad \frac{\partial E}{\partial(\Phi/\Phi_0)} = \frac{2\pi \hbar c}{L}. \quad (11.6)$$

Topological mass floors. By Fenchel ($\oint_{\Gamma} |\kappa_g| ds \geq 2\pi$) and Fáry–Milnor ($\geq 4\pi$ for knots),

$$m \geq \frac{\hbar}{c} \frac{2\pi}{L} \quad (\textit{unknotted}), \quad m \geq \frac{\hbar}{c} \frac{4\pi}{L} \quad (\textit{knotted}). \quad (11.7)$$

12 Lyapunov descent and Gibbs form

The functional derivative of (10.1) is

$$\frac{\delta F}{\delta \phi} = -\hbar c \partial_s [w(\partial_s \phi - \sigma)]. \quad (12.1)$$

Deterministic gradient flow $\partial_t \phi = -\mu \delta F / \delta \phi$ yields

$$\frac{dF}{dt} = \int \frac{\delta F}{\delta \phi} \partial_t \phi \, ds = -\mu \int \left| \frac{\delta F}{\delta \phi} \right|^2 ds \leq 0. \quad (12.2)$$

With additive white noise, $\partial_t \phi = -\mu \delta F / \delta \phi + \sqrt{2\mu k_B T_{\text{eff}}} \xi$, the stationary measure is Gibbs:

$$\mathbb{P}[\phi] \propto \exp(-\beta_{\text{eff}} F[\phi]), \quad \beta_{\text{eff}} = (k_B T_{\text{eff}})^{-1}. \quad (12.3)$$

Remark (media). In a medium of phase speed v_{ph} , replace $c \rightarrow v_{\text{ph}}$ in (10.1), (11.4), and (11.6).

13 Worked example (one number)

Circle $R = 1 \text{ cm} \Rightarrow L = 2\pi R$, $\kappa_g = 1/R$, $\tau = K_{\text{ext}} = 0$. At lock with $n = 1$, $\Phi = 0$, $S\Gamma_s = 0$:

$$k_{\text{eff}} = \langle \sigma \rangle_w = \frac{1}{R} = 100 \text{ m}^{-1}, \quad E = \hbar c k_{\text{eff}} \approx 3.16 \times 10^{-24} \text{ J} (\approx 1.97 \times 10^{-5} \text{ eV}), \quad m = \frac{\hbar k_{\text{eff}}}{c} \approx 3.52 \times 10^{-41} \text{ kg}. \quad (13.1)$$

Definition 13.1 (Rest mass via curvature lock). In vacuum and in the comoving rest slice, if the loop is locked ($\alpha = 0$), the rest mass is

$$m := \frac{\hbar}{c} \langle \sigma \rangle_w = \frac{\hbar}{c} \frac{B}{L},$$

so that $E_{\text{rest}} = mc^2 = \hbar c B/L$.

Proposition 7 (General inertial mass parameter). *For arbitrary mislock or media,*

$$m_{\text{gen}} := \frac{\hbar}{c} \langle \partial_s \phi \rangle_w = \frac{\hbar}{c} \left[\langle \sigma \rangle_w + \frac{L}{\oint w ds} \cdot \frac{B - \oint \sigma ds}{\oint w^{-1} ds} \right].$$

Here the deviation $m_{\text{gen}} - m$ (when defined) is controlled by the conserved mislock α and vanishes at lock.

Remark 2 (Interpretation). B encodes topological/gauge holonomy (“photon relationships”); α encodes inertial torque (misfit). The law above therefore describes mass as holonomy per length, corrected by torque when off-lock.

Conclusion

$$m = \frac{\hbar}{c} \langle \sigma \rangle_w, \quad E_{\text{rest}} = \hbar c \langle \sigma \rangle_w$$

follows from the locking principle on Γ , admits exact holonomy and topology corrections, and is supported by strict stability plus quantitative mislock bounds suitable for calibration against experiment.

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A Ring model on S^1 : symbols and equation

Symbols. s = arclength on a circle of radius $R > 0$; domain $s \in [0, 2\pi R)$. $\psi = \psi(s, t) \in \mathbb{C}$ (field). Parameters: dispersion $\kappa > 0$, cubic nonlinearity $g \in \mathbb{R}$, pump $\mu > 0$, linear loss $\gamma > 0$, saturating loss $\eta > 0$, rotation rate $\Omega \in \mathbb{R}$. Fourier modes: $m \in \mathbb{Z}$, basis $e_m(s) = \frac{1}{\sqrt{2\pi R}} e^{ims/R}$, with $\partial_s e_m = i(m/R)e_m$ and $\partial_{ss} e_m = -(m/R)^2 e_m$. Inner product on $L^2(S^1)$: $\langle f, g \rangle := \int_0^{2\pi R} f^* g ds$.

Equation (CGLE on S^1).

$$i \partial_t \psi = -\kappa \partial_{ss} \psi + g |\psi|^2 \psi - \Omega (-i) \partial_s \psi + i(\mu - \gamma - \eta |\psi|^2) \psi. \quad (\text{A.1})$$

B Linear spectrum and saturated rotating waves

Proposition 8 (Linear band and growth rates). *Let $\psi(s, t) = \epsilon e^{i(ms/R - \omega t)}$ with $\epsilon \rightarrow 0$. Linearizing (D.1) about $\psi = 0$ yields the modal growth rate*

$$\Re \lambda_m = \mu - \gamma - \kappa(m/R)^2,$$

so the linearly unstable band is

$$|m| < R \sqrt{\frac{\mu - \gamma}{\kappa}}.$$

Proof. Insert $\psi = \epsilon e^{i(ms/R - \omega t)}$: $\partial_t \psi = -i\omega \psi$, $\partial_s \psi = i(m/R)\psi$, $\partial_{ss} \psi = -(m/R)^2 \psi$. Neglecting $O(\epsilon^3)$ terms (nonlinearity and saturation), (D.1) gives

$$i(-i\omega)\psi = \kappa(m/R)^2 \psi + \Omega(m/R)\psi + i(\mu - \gamma)\psi.$$

Cancel $\psi \neq 0$: $\omega = \kappa(m/R)^2 + \Omega(m/R) + i(\mu - \gamma)$, hence $\Re \lambda_m = \mu - \gamma - \kappa(m/R)^2$. Unstable modes have $\Re \lambda_m > 0$. $\square$

Proposition 9 (Nonlinear saturated rotating wave). *Seek a time-periodic single-mode state $\psi(s, t) = A e^{i(ms/R - \omega t)}$ with $A \in \mathbb{C}$. Then $|A|^2 = (\mu - \gamma)/\eta$ (for $\mu > \gamma$) and*

$$\omega = \kappa(m/R)^2 + g|A|^2 - \Omega m/R.$$

Proof. Write $\rho := |A|^2$. Substitute $\psi = A e^{i(ms/R - \omega t)}$ into (D.1) including the cubic and saturation terms:

$$\omega = \kappa(m/R)^2 + g\rho - \Omega m/R + i(\mu - \gamma - \eta\rho).$$

For a genuine periodic solution the imaginary part must vanish: $\mu - \gamma - \eta\rho = 0 \Rightarrow \rho = (\mu - \gamma)/\eta$. The real part yields the stated ω . $\square$

Observable. The phase $S = ms/R - \omega t$ has tangential wavenumber $k_\theta = m/R$, hence

$$v_\theta := \frac{\partial \omega}{\partial k_\theta} \text{ (plateau velocity)} \equiv \frac{m}{R}$$

for the locked integer m (units scaled so the advective term is $-\Omega(-i)\partial_s$).

C Modal equations and single-mode capture (no sketch)

Fourier expansion and averaging. Expand $\psi(s, t) = \sum_m a_m(t) e_m(s)$ with $a_m \in \mathbb{C}$ and define modal intensities $x_m := |a_m|^2$. Project (D.1) onto e_m using $\langle e_\ell, e_m \rangle = \delta_{\ell m}$. The linear and advective parts contribute $(\kappa(m/R)^2 + \Omega(m/R) + i(\mu - \gamma))a_m$ as in Prop. 8. For the cubic terms we employ the rotating-wave/mean-field approximation on the ring:

$$|\psi|^2 \psi = \left(\sum_j |a_j|^2 \right) \left(\sum_m a_m e_m \right) + \text{nonresonant beats},$$

and the nonresonant beats average to zero on S^1 over time (orthogonality of unequal harmonics). Thus the cubic conservative term projects to $gS a_m$ and the saturating term to $-i\eta S a_m$, where

$$S := \sum_{n \in \mathbb{Z}} x_n \quad (\text{total modal intensity}).$$

We obtain the closed modal system

$$\dot{a}_m = \left[-i(\kappa(m/R)^2 - \Omega m/R - gS) + (\mu - \gamma - \eta S) \right] a_m. \quad (\text{C.1})$$

Intensity dynamics. Taking $x_m = |a_m|^2$ and differentiating,

$$\dot{x}_m = \dot{a}_m a_m^* + a_m \dot{a}_m^* = 2(\mu - \gamma - \eta S) x_m.$$

Hence the *coupled* intensity equations reduce to

$$\dot{x}_m = 2(\lambda_m - \eta S) x_m, \quad \lambda_m := \mu - \gamma - \kappa(m/R)^2. \quad (\text{C.2})$$

Theorem C.1 (Single-mode capture under shared saturation). *Assume the growth rates λ_m achieve a unique maximum at m^* : $\lambda_{m^*} > \lambda_m$ for all $m \neq m^*$. Then for any initial data $x_m(0) \geq 0$ with $S(0) > 0$, the solution of (C.2) satisfies*

$$x_{m^*}(t) \rightarrow \frac{\lambda_{m^*}}{\eta}, \quad x_m(t) \rightarrow 0 \quad (m \neq m^*), \quad S(t) \rightarrow \frac{\lambda_{m^*}}{\eta} \quad \text{as } t \rightarrow \infty.$$

In particular, the only asymptotically stable equilibrium in the positive cone is the single-mode state with the maximally unstable m^ .*

Proof. Step 1: positivity and invariance. From (C.2), if $x_m(0) \geq 0$ then $x_m(t) \geq 0$ ($x_m = 0$ is invariant). Let $S = \sum_n x_n$.

Step 2: ratio dynamics. For any $m \neq m^*$ with $x_m, x_{m^*} > 0$, compute

$$\frac{d}{dt} \ln \left(\frac{x_m}{x_{m^*}} \right) = \frac{\dot{x}_m}{x_m} - \frac{\dot{x}_{m^*}}{x_{m^*}} = 2(\lambda_m - \eta S - (\lambda_{m^*} - \eta S)) = 2(\lambda_m - \lambda_{m^*}) < 0.$$

Thus $\frac{x_m(t)}{x_{m^*}(t)} \leq C_m e^{-2(\lambda_{m^*} - \lambda_m)t} \rightarrow 0$ exponentially.

Step 3: limit of S . Sum (C.2) over m :

$$\dot{S} = 2 \sum_m \lambda_m x_m - 2\eta S^2 = 2\lambda_{m^*} x_{m^*} + 2 \sum_{m \neq m^*} \lambda_m x_m - 2\eta S^2.$$

Using Step 2, $x_m = o(x_{m^*})$ for $m \neq m^*$, hence $\sum_{m \neq m^*} \lambda_m x_m = o(x_{m^*})$, and $S = x_{m^*} + o(x_{m^*})$. Therefore,

$$\dot{S} = 2\lambda_{m^*} S - 2\eta S^2 + o(S).$$

Comparison with the scalar logistic equation $\dot{y} = 2\lambda_{m^*} y - 2\eta y^2$ implies $S(t) \rightarrow \lambda_{m^*}/\eta$.

Step 4: component limits. From Step 2 and $S \rightarrow \lambda_{m^*}/\eta$ we have $x_{m^*} = S - \sum_{m \neq m^*} x_m \rightarrow \lambda_{m^*}/\eta$, while each $x_m \rightarrow 0$ for $m \neq m^*$. Thus the unique asymptotically stable equilibrium is the single-mode state. $\square$

Rotation frequency at capture. At the captured equilibrium $S = \lambda_{m^*}/\eta$, the phase equation from (C.1) gives

$$\omega = \kappa(m^*/R)^2 + g \frac{\lambda_{m^*}}{\eta} - \Omega m^*/R.$$

D Lyapunov/enthalpy descent (strict inequality)

Setup. On the ring $\Gamma \simeq S^1$ with arclength $s \in [0, L_s)$ (periodic), consider the CGLE-type evolution

$$\psi_t = -i \left(-\kappa \psi_{ss} + g|\psi|^2 \psi + i \Omega \psi_s \right) + (\mu - \gamma - \eta|\psi|^2) \psi, \quad (\text{D.1})$$

with real constants $\kappa, g, \Omega, \mu, \gamma, \eta$. Write $\psi = \sqrt{\rho} e^{i\theta}$ so that

$$\rho_t = 2\kappa \partial_s(\rho \theta_s) + \Omega \rho_s + 2(\mu - \gamma - \eta\rho) \rho. \quad (\text{D.2})$$

Definition D.1 (Enthalpy density and functional). Let $\rho = |\psi|^2$. Define

$$h(\rho) := \frac{\eta}{2} \rho^2 - (\mu - \gamma) \rho, \quad \mathcal{H}[\psi] := \int_0^{L_s} h(|\psi(s, t)|^2) ds.$$

Proposition 10 (Exact balance law). *Along any classical solution of (D.1),*

$$\frac{d\mathcal{H}}{dt} = - \int_0^{L_s} (\mu - \gamma - \eta\rho)^2 2\rho ds - 2\kappa\eta \int_0^{L_s} \rho_s \rho \theta_s ds. \quad (\text{D.3})$$

The advection term $(\Omega \rho_s)$ contributes only a periodic boundary flux and cancels: $\int h'(\rho) \Omega \rho_s ds = \Omega [h(\rho)]_0^{L_s} = 0$.

Proof. Differentiate $\mathcal{H}$ and use (D.2):

$$\frac{d\mathcal{H}}{dt} = \int h'(\rho) \rho_t ds = \int h'(\rho) (2\kappa(\rho \theta_s)_s + \Omega \rho_s + 2(\mu - \gamma - \eta\rho)\rho) ds.$$

The Ω -term vanishes by periodicity. For the κ -term, integrate by parts once and use $h''(\rho) = \eta$:

$$\int h'(\rho) 2\kappa(\rho \theta_s)_s ds = -2\kappa \int h''(\rho) \rho_s \rho \theta_s ds = -2\kappa\eta \int \rho_s \rho \theta_s ds.$$

The remaining term gives $-\int 2(\mu - \gamma - \eta\rho)^2 \rho ds$. Summing yields (D.3). $\square$

Corollary 7 (Strict descent under natural conditions). *The functional $\mathcal{H}$ is strictly decreasing except at equilibria in any of the following regimes:*

1. **No Laplacian drive:** $\kappa = 0$ (pure advection + local gain/loss).
2. **Locked or uniform amplitude:** $\rho_s \equiv 0$ (e.g. on the locked manifold after amplitude saturation).
3. **Dominant dissipation:** for general $\kappa \neq 0$, if

$$2\kappa\eta \left| \int \rho_s \rho \theta_s ds \right| \leq \int 2(\mu - \gamma - \eta\rho)^2 \rho ds,$$

then $\frac{d\mathcal{H}}{dt} \leq 0$, with equality iff $\mu - \gamma - \eta\rho \equiv 0$ a.e.

Remark 3 (Interpretation). *The first integral on the right of (D.3) is the local enthalpy dissipation from amplitude relaxation; the second is a Hamiltonian coupling flux produced by curvature of the phase (θ_s) against amplitude gradients (ρ_s). It vanishes identically when $\kappa = 0$, when the amplitude is spatially uniform, or asymptotically after locking when $\rho \rightarrow \rho_* := (\mu - \gamma)/\eta$.*

E Rest scale via constrained variational matching

Data. Let $\sigma = \sigma(s) > 0$ be a prescribed target phase gradient, $w = w(s) > 0$ a weight, and $L_s := \int_0^{2\pi R} ds = 2\pi R$ the loop length.

Definition E.1 (Rest scale). For $n \in \mathbb{Z}$, define

$$k_{\text{eff}} := \arg \min_{\phi \in H^1(S^1)} \oint w(s)(\phi'(s) - \sigma(s))^2 ds \quad \text{subject to} \quad \oint \phi'(s) ds = 2\pi n, \quad E_{\text{rest}} := \hbar c k_{\text{eff}}. \quad (\text{E.1})$$

Theorem E.2 (Euler–Lagrange, uniqueness, bounds). *There exists a unique minimizer modulo constants. It satisfies*

$$\phi'(s) = \sigma(s) - \frac{\Lambda}{2w(s)}, \quad \Lambda = \frac{2}{\oint w^{-1} ds} \left(\oint \sigma ds - 2\pi n \right), \quad (\text{E.2})$$

and the optimal effective wavenumber on branch n is

$$k_{\text{eff}} = \frac{2\pi n}{L_s} = \frac{\oint w \sigma ds}{\oint w ds} = \langle \sigma \rangle_w.$$

Moreover, for $w > 0$ and $\sigma > 0$,

$$\sigma_{\min} \leq k_{\text{eff}} = \langle \sigma \rangle_w \leq \sigma_{\max}, \quad \frac{1}{\langle \sigma^{-1} \rangle_w} \leq k_{\text{eff}}.$$

Proof. Consider the Lagrangian functional $\mathcal{J}[\phi] = \oint w(\phi' - \sigma)^2 ds + \Lambda(\oint \phi' ds - 2\pi n)$. Variations $\phi \mapsto \phi + \epsilon \eta$ with periodic η give

$$\delta \mathcal{J} = \oint (2w(\phi' - \sigma) + \Lambda) \eta' ds = - \oint \partial_s (2w(\phi' - \sigma) + \Lambda) \eta ds.$$

Since η is arbitrary, $\partial_s (2w(\phi' - \sigma) + \Lambda) = 0$, so $2w(\phi' - \sigma) + \Lambda \equiv C$, a constant. Periodic boundary plus the constraint fixes $C = 0$ (any nonzero C would violate the constraint for general w), hence $\phi' = \sigma - \Lambda/(2w)$.

Integrate ϕ' on S^1 : $2\pi n = \oint \phi' ds = \oint \sigma ds - \frac{\Lambda}{2} \oint w^{-1} ds$, which gives the stated Λ .

Uniqueness modulo constants: The integrand $w(\phi' - \sigma)^2$ is strictly convex in ϕ' , and the constraint is affine; therefore the minimizer is unique up to an additive constant in ϕ .

Value of k_{eff} : The total phase advance is $2\pi n = \oint \phi' ds$. Define the effective wavenumber by $k_{\text{eff}} := \frac{1}{L_s} \oint \phi' ds$. Then $k_{\text{eff}} = 2\pi n / L_s$. On the other hand, insert $\phi' = \sigma - \Lambda/(2w)$ and the expression for Λ to get

$$\oint \phi' ds = \oint \sigma ds - \frac{\Lambda}{2} \oint w^{-1} ds = \oint \sigma ds - \left(\oint \sigma ds - 2\pi n \right) = 2\pi n,$$

while the stationarity condition can be written equivalently as $\oint w(\phi' - \sigma) ds = 0 \Rightarrow \oint w \phi' ds = \oint w \sigma ds$. Divide by $\oint w ds$ to obtain $k_{\text{eff}} = \langle \sigma \rangle_w$.

Bounds: From positivity of w , $\sigma_{\min} \leq \langle \sigma \rangle_w \leq \sigma_{\max}$ is immediate. For the harmonic-type bound, by Cauchy–Schwarz on S^1 with weight w ,

$$\left(\oint w \sigma ds \right) \left(\oint w \sigma^{-1} ds \right) \geq \left(\oint w ds \right)^2,$$

so $\langle \sigma \rangle_w \langle \sigma^{-1} \rangle_w \geq 1$, i.e. $1/\langle \sigma^{-1} \rangle_w \leq \langle \sigma \rangle_w = k_{\text{eff}}$. □

F Gauge and spin holonomy shifts (explicit)

Coupling. Let $A_s(s) := \mathbf{A} \cdot \hat{\mathbf{t}}$ (tangential vector potential) and $\omega_s(s)$ the spin connection along the loop. Let q be charge and s the spin projection (e.g. $s = \pm \frac{1}{2}$ for spin- $\frac{1}{2}$).

Corollary 8 (Holonomy shift). *Modify the functional in (E.1) by replacing $\phi'(s)$ with $\phi'(s) + \frac{q}{\hbar} A_s(s) + s \omega_s(s)$ and impose the constraint on the gauge/spin covariant phase advance:*

$$\oint \left(\phi' + \frac{q}{\hbar} A_s + s \omega_s \right) ds = 2\pi n.$$

Then the minimizer is as in Theorem E.2 with $\sigma \mapsto \sigma + \frac{q}{\hbar} A_s + s \omega_s$, yielding

$$k_{\text{eff}} = \frac{2\pi}{L_s} \left(n + \frac{\Phi}{\Phi_0} + s \Gamma_s \right), \quad \Phi := \oint \mathbf{A} \cdot d\mathbf{l}, \quad \Phi_0 := \frac{h}{q}, \quad \Gamma_s := \frac{1}{2\pi} \oint \omega_s ds. \quad (\text{F.1})$$

Proof. Define $\tilde{\sigma} := \sigma + \frac{q}{\hbar} A_s + s \omega_s$. Repeat the Euler–Lagrange steps of Theorem E.2 with σ replaced by $\tilde{\sigma}$. Then $k_{\text{eff}} = \langle \tilde{\sigma} \rangle_w$ and, evaluating the loop averages,

$$\langle \tilde{\sigma} \rangle_w = \langle \sigma \rangle_w + \frac{q}{\hbar} \frac{\oint A_s ds}{L_s} + s \frac{\oint \omega_s ds}{L_s} = \frac{2\pi n}{L_s} + \frac{2\pi}{L_s} \left(\frac{\Phi}{\Phi_0} + s \Gamma_s \right).$$

□

Spin factor convention. For spin- $\frac{1}{2}$, the geometric phase is carried by $s = \pm \frac{1}{2}$. Alternative conventions absorb the 1/2 into ω_s ; (F.1) remains unchanged.

G Mass–Energy from Curvature Locking

Definitions (fixed).

- $R > 0$: ring radius; $L_s = 2\pi R$.
- $\ell \in \mathbb{Z}$: winding; Φ : magnetic flux; $\Phi_0 = h/q$: flux quantum.
- s : spin projection (e.g. $s = \pm \frac{1}{2}$); $\Gamma_s = \frac{1}{2\pi} \oint \omega_s ds$: spin holonomy.
- $\theta := \ell + \frac{\Phi}{\Phi_0} + s \Gamma_s$ (dimensionless holonomy index).
- $k_{\text{eff}} := \frac{\theta}{R}$ (locked wavenumber; units m^{-1}).
- $P := \hbar k_{\text{eff}}$ (curvature momentum scale; units $\text{kg} \cdot \text{m/s}$).
- c : causal speed invariant; $p := \hbar k$ (linear momentum).

Core identities.

$$E_{\text{rest}} = cP = \hbar c k_{\text{eff}} = \frac{\hbar c}{R} \theta, \quad m = \frac{P}{c} = \frac{\hbar}{c} k_{\text{eff}} = \frac{\hbar}{cR} \theta, \quad \frac{E_{\text{rest}}}{m} = c^2.$$

Relativistic dispersion (derivation). Starting with $p = \hbar k$ and $m = \frac{\hbar}{c} k_{\text{eff}}$,

$$E^2 = (pc)^2 + (mc^2)^2 = (\hbar k c)^2 + \left(\frac{\hbar}{c} k_{\text{eff}} c^2\right)^2 = (\hbar c)^2 (k^2 + k_{\text{eff}}^2). \quad (\text{G.1})$$

Rest limit $k = 0$ gives $E_{\text{rest}} = \hbar c k_{\text{eff}}$. Low- k expansion:

$$E = \hbar c \sqrt{k^2 + k_{\text{eff}}^2} = \hbar c k_{\text{eff}} + \frac{\hbar c}{2k_{\text{eff}}} k^2 + \mathcal{O}(k^4) = E_{\text{rest}} + \frac{p^2}{2m} + \mathcal{O}(k^4). \quad (\text{G.2})$$

Level spacing and slopes (invariants).

$$\Delta E = E_{\ell+1} - E_{\ell} = \frac{\hbar c}{R}, \quad \frac{\partial E}{\partial \Phi} = \frac{q c}{L_s}, \quad L_z = \ell \hbar, \quad \frac{\partial E}{\partial L_z} = \frac{c}{R}.$$

Worked example (numbers). For $R = 5 \text{ mm}$, $L_s \approx 3.1416 \times 10^{-2} \text{ m}$, $\Phi = \Gamma_s = 0$:

$$\Delta E = \frac{\hbar c}{R} \approx 6.32 \times 10^{-24} \text{ J}, \quad \Delta f = \frac{\Delta E}{h} = \frac{c}{L_s} \approx 9.54 \text{ GHz}, \quad \Delta m = \frac{\hbar}{cR} \approx 7.03 \times 10^{-41} \text{ kg}.$$

Interpretation (concise). Mass is the curvature/holonomy index scaled by $\hbar/c$; energy is the same index scaled by $\hbar c$. Geometry/holonomy (R, Φ, Γ_s) move (k_{eff}, m, E) ; pump/loss parameters only set amplitude/lifetime and do not shift E_{rest} within the locked basin.

H Variables and Units (Gravity/Relativity Addendum)

- Geometry: $R > 0$ [m] (ring radius), $L_s := 2\pi R$ [m] (circumference), $s \in [0, L_s]$ [m] (arclength).
- Integers/holonomy: $\ell \in \mathbb{Z}$ (winding), Φ [Wb] (flux), q [C] (charge), $\Phi_0 := h/q$ [Wb] (flux quantum), s (spin projection), $\omega_s(s)$ [rad/m] (spin connection), $\Gamma_s := \frac{1}{2\pi} \oint \omega_s ds$ (spin holonomy), $\theta := \ell + \Phi/\Phi_0 + s \Gamma_s$ (holonomy index).
- Field/weights: $\psi(s, t) \in \mathbb{C}$, $\rho := |\psi|^2$ (intensity), $\phi(s)$ [rad] (phase), $\sigma := \partial_s \phi$ [m⁻¹] (phase gradient), $w(s) > 0$ [J/m] (line enthalpy), $\langle f \rangle_w := \frac{\oint w f ds}{\oint w ds}$.
- Kinematics: k [m⁻¹] (small wave), $k_{\text{eff}} := \langle \sigma \rangle_w = \theta/R$ [m⁻¹], $p := \hbar k$ [kg m/s], $P := \hbar c k_{\text{eff}}$ [kg m/s], $m := \hbar c k_{\text{eff}}/c$ [kg], E [J], $f := E/h$ [Hz].
- Gravity/relativity: $T_{\mu\nu}$ [J/m³] (stress-energy), $T_{00} \equiv u(\mathbf{x})$ (energy density), G [m³kg⁻¹s⁻²], c [m/s], $\phi(\mathbf{x}, t)$ [m²/s²] (Newtonian potential), $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$, $\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, $h := \eta^{\alpha\beta}h_{\alpha\beta}$, $\square := -c^{-2}\partial_t^2 + \nabla^2$.

I Mass–Energy from Curvature Locking (Recall)

$$k_{\text{eff}} = \frac{\theta}{R} = \langle \sigma \rangle_w, \quad P = \hbar c k_{\text{eff}}, \quad m = \frac{P}{c} = \frac{\hbar}{c} k_{\text{eff}}, \quad E_{\text{rest}} = cP = \hbar c k_{\text{eff}} = \frac{\hbar c}{R} \theta, \quad (\text{I.1})$$

$$E^2 = (pc)^2 + (mc^2)^2 = (\hbar c)^2 (k^2 + k_{\text{eff}}^2), \quad E = E_{\text{rest}} + \frac{p^2}{2m} + \mathcal{O}(k^4). \quad (\text{I.2})$$

$$\Delta E = \frac{\hbar c}{R}, \quad \frac{\partial E}{\partial \Phi} = \frac{q c}{L_s}, \quad L_z = \ell \hbar \Rightarrow \frac{\partial E}{\partial L_z} = \frac{c}{R}. \quad (\text{I.3})$$

J Newtonian Coupling from an Action (G1)

Definition J.1 (Newtonian sector Lagrangian).

$$\mathcal{L}_\phi = +\frac{c^2}{8\pi G} (\nabla\phi)^2 + \frac{\phi}{c^2} T_{00}. \quad (\text{J.1})$$

Proposition 11 (Poisson equation). *The Euler–Lagrange equation for ϕ implied by $\mathcal{L}_\phi$ is*

$$\nabla^2\phi = \frac{4\pi G}{c^2} T_{00}. \quad (\text{J.2})$$

Proof. $\partial\mathcal{L}_\phi/\partial\phi = T_{00}/c^2$, $\partial\mathcal{L}_\phi/\partial(\partial_i\phi) = \frac{c^2}{4\pi G} \partial_i\phi$. Euler–Lagrange: $\frac{T_{00}}{c^2} - \partial_i(\frac{c^2}{4\pi G} \partial_i\phi) = 0$. $\square$

Corollary 9 (EP and redshift). *For any locked object with $E_{\text{rest}}^0 = mc^2$,*

$$E_{\text{rest}}(\mathbf{x}) = E_{\text{rest}}^0 \left(1 + \frac{\phi(\mathbf{x})}{c^2}\right), \quad \mathbf{F} = -\nabla E_{\text{rest}} = -m \nabla\phi, \quad \frac{\Delta f}{f} = \frac{\Delta\phi}{c^2}. \quad (\text{J.3})$$

Thus free-fall acceleration $\mathbf{a} = \mathbf{F}/m = -\nabla\phi$ is composition-independent.

K Linearized Spin-2 Sector and $\gamma = 1$ (G2)

Definition K.1 (Weak-field metric response). Choose

$$g_{tt} = -(1 + 2\phi/c^2), \quad g_{ij} = (1 - 2\phi/c^2)\delta_{ij}, \quad (\text{K.1})$$

equivalently $h_{00} = -2\phi/c^2$, $h_{ij} = -2\phi/c^2 \delta_{ij}$.

Proposition 12 (Linearized Einstein equations). *In harmonic gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$,*

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}. \quad (\text{K.2})$$

Static limit ($\partial_t = 0$) and $T_{00} \gg T_{ij}$ give $\nabla^2 \bar{h}_{00} = -16\pi G T_{00}/c^4$. With $h = -h_{00} + h_{ii} = 4\phi/c^2$ and $\bar{h}_{00} = h_{00} - \frac{1}{2}\eta_{00}h = -2\phi/c^2 + \frac{1}{2}(4\phi/c^2) = -4\phi/c^2$,

$$\nabla^2(-4\phi/c^2) = -\frac{16\pi G}{c^4} T_{00} \Rightarrow \nabla^2\phi = \frac{4\pi G}{c^2} T_{00}. \quad (\text{K.3})$$

Corollary 10 (PPN parameter). *Since the spatial and temporal coefficients are equal in magnitude, $\gamma = 1$; hence light bending and Shapiro delay match GR at leading order.*

L Gravitational Waves (G3)

Proposition 13 (Vacuum waves and polarizations). *For $T_{\mu\nu} = 0$, $\square \bar{h}_{\mu\nu} = 0$ supports plane waves with $\omega = ck$. In transverse-traceless gauge, two physical polarizations $(+, \times)$ propagate at c .*

M $\psi \rightarrow$ Metric/Source Map (G4)

Definition M.1 (Effective action and universal coupling). Let $S_{\text{eff}}[\psi, g]$ be the ψ -sector action on (M, g) ; define

$$T_{\mu\nu}(\psi, g) := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{eff}}}{\delta g^{\mu\nu}}, \quad \text{and ensure } E_{\text{rest}} = \hbar c \langle \sigma \rangle_w. \quad (\text{M.1})$$

The linearized gravitational sector couples universally as $\frac{1}{2}h_{\mu\nu}T^{\mu\nu}$.

N Locking Time Bound (G5)

Setup. Let mode amplitudes $a_m(t) \in \mathbb{C}$ with powers $x_m := |a_m|^2 \geq 0$ and

$$\dot{x}_m = 2(\lambda_m - \eta S) x_m, \quad S := \sum_n x_n, \quad \lambda_{m^*} = \max_m \lambda_m, \quad \lambda_{(2)} := \max_{m \neq m^*} \lambda_m, \quad (\text{N.1})$$

with $\eta > 0$ and a strict spectral gap $\Delta := \lambda_{m^*} - \lambda_{(2)} > 0$.

Lemma N.1 (Exponential separation by spectral gaps). *For $m \neq m^*$,*

$$\frac{d}{dt} \ln \frac{x_m}{x_{m^*}} = 2(\lambda_m - \lambda_{m^*}) \implies \frac{x_m(t)}{x_{m^*}(t)} = \frac{x_m(0)}{x_{m^*}(0)} e^{-2(\lambda_{m^*} - \lambda_m)t}. \quad (\text{N.2})$$

Hence

$$\sum_{m \neq m^*} \frac{x_m(t)}{x_{m^*}(t)} \leq \left(\sum_{m \neq m^*} \frac{x_m(0)}{x_{m^*}(0)} \right) e^{-2\Delta t}. \quad (\text{N.3})$$

Corollary 11 (Capture time at fidelity threshold). *Let the target fidelity be $p_*(t) := \frac{x_{m^*}(t)}{S(t)} \geq 1 - \varepsilon$ with $\varepsilon \in (0, 1)$. A sufficient locking time is*

$$t_{\text{lock}}(\varepsilon) \leq \frac{1}{2\Delta} \ln \left(\frac{1}{\varepsilon} \frac{\sum_{m \neq m^*} x_m(0)}{x_{m^*}(0)} \right), \quad (\text{N.4})$$

because $p_* = (1 + \sum_{m \neq m^*} x_m/x_{m^*})^{-1} \geq (1 + \varepsilon)^{-1} \geq 1 - \varepsilon$ when (N.3) $\leq \varepsilon$.

Remark 4 (Tightness and invariance). *The bound depends only on the spectral gap Δ and the initial power ratios; the nonlinear common loss $-\eta S$ cancels in (N.2), making (N.4) gap-optimal for this model.*

O Function-Space Assumptions and Bounds (G6)

Assumption O.1 (Well-posed variational class). $\Gamma \subset \mathbb{R}^3$ is C^2 and closed; $\sigma \in L^2(\Gamma)$; $w \in L^\infty(\Gamma)$ with $w \geq w_{\min} > 0$ a.e. Phases $\phi \in H^1(S^1)$ satisfy the winding constraint $\oint_\Gamma \phi' ds = 2\pi n$. Consider

$$\min_{\phi \in H^1} \mathcal{J}[\phi] := \oint_\Gamma w(s) (\phi'(s) - \sigma(s))^2 ds \quad \text{s.t.} \quad \oint_\Gamma \phi'(s) ds = 2\pi n.$$

Proposition 14 (Strict convexity, existence/uniqueness). *$\mathcal{J}$ is strictly convex on the affine subspace defined by the linear constraint; therefore a unique minimizer exists modulo additive constants. Its Euler-Lagrange equation is*

$$\partial_s(2w(\phi' - \sigma) + \Lambda) = 0 \implies \phi' = \sigma + \frac{\alpha}{w}, \quad \alpha = \frac{2\pi n - \oint_\Gamma \sigma ds}{\oint_\Gamma \frac{ds}{w}}.$$

Perfect lock $\phi' = \sigma$ holds iff $\oint_\Gamma \sigma ds = 2\pi n$.

Proposition 15 (Estimator bounds (HM-AM- ∞ chain)). *Assume $\sigma > 0$ a.e. and $w > 0$. Then*

$$\frac{1}{\langle \sigma^{-1} \rangle_w} = \frac{\oint_\Gamma w ds}{\oint_\Gamma \frac{w}{\sigma} ds} \leq \langle \sigma \rangle_w \leq \|\sigma\|_{L^\infty(\Gamma)}. \quad (\text{O.1})$$

In particular, under perfect lock, $k_{\text{eff}} = \langle \sigma \rangle_w$ is bounded between the weighted harmonic and the essential supremum of σ .

Proposition 16 (Mass/estimator error at the minimizer). *Let $\delta := \phi' - \sigma = \alpha/w$ at the minimizer. Then*

$$\left| k_{\text{eff}} - \langle \sigma \rangle_w \right| = \langle \delta \rangle_w \leq \left(\frac{\oint_{\Gamma} w \delta^2 ds}{\oint_{\Gamma} w ds} \right)^{1/2}, \quad \left| m - \frac{\hbar}{c} \langle \sigma \rangle_w \right| \leq \frac{\hbar}{c} \left(\frac{\oint_{\Gamma} w \delta^2 ds}{\oint_{\Gamma} w ds} \right)^{1/2}. \quad (\text{O.2})$$

Sketches. *Prop. 14:* quadratic $\mathcal{J}$ with $w \geq w_{\min} > 0$ is strictly convex on the affine subspace; apply Lax–Milgram on $H^1/\mathbb{R}$. *Prop. 15:* weighted HM $\leq$ AM and AM $\leq \|\cdot\|_{\infty}$. *Prop. 16:* Cauchy–Schwarz with weight w . $\square$

P Worked Example (Numbers; Instantiation)

Example P.1 (Quantized ring step ($n \rightarrow n+1$) and gravity scale). Let $R = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$, so $L_s = 2\pi R \approx 3.14 \times 10^{-2} \text{ m}$. Take $\Phi = \Gamma_s = 0$ and winding $n = 1$. Then $k_{\text{eff}} = 1/R = 200 \text{ m}^{-1}$ and

$$E_{\text{rest}} = \hbar c k_{\text{eff}} = \frac{\hbar c}{R} \approx 6.32 \times 10^{-24} \text{ J} (= 3.94 \times 10^{-5} \text{ eV}), \quad \Delta f = \frac{c}{L_s} \approx 9.55 \text{ GHz}, \quad m = \frac{\hbar}{cR} \approx 7.03 \times 10^{-41} \text{ kg}. \quad (\text{P.1})$$

Consistency check:

$$E_{\text{rest}} = h \Delta f$$

within rounding since $k_{\text{eff}} = 2\pi/L_s$.

For an energy density $u = T_{00} = 10^3 \text{ J m}^{-3}$ localized on scale $L \sim 1 \text{ m}$, the Newtonian limit $\nabla^2 \phi = \frac{4\pi G}{c^2} u$ gives the characteristic curvature

$$\|\nabla^2 \phi\| \approx \frac{4\pi G}{c^2} u \approx 9.33 \times 10^{-24} \text{ s}^{-2},$$

deep in the linear regime; metric perturbations satisfy $|h_{\mu\nu}| \sim |\phi|/c^2 \ll 1$, and GW propagation obeys $\omega = ck$.

Q Closure: From WCT to Gravity (Action–Level Derivation)

Q.1 Fixed primitives and notation

- Spacetime: $(M, g_{\mu\nu})$, signature $(-, +, +, +)$, $\eta_{\mu\nu} = \text{diag}(-, +, +, +)$, c the causal speed.
- WCT field: ψ (complex), $\rho := |\psi|^2$, phase ϕ , phase gradient $\Sigma_{\mu} := \nabla_{\mu} \phi$.
- Covariant WCT action (generic two-derivative core; higher-order terms absorbed in $\mathcal{U}$):

$$S_{\text{WCT}}[\psi, g] = \int d^4x \sqrt{-g} \left(g^{\mu\nu} \nabla_{\mu} \psi \nabla_{\nu} \bar{\psi} - \mathcal{U}(\rho, \Sigma^2; g) \right).$$

$\mathcal{U}$ collects nonlinearity, locking feedback, and regularizers already introduced in the ring model.

- Stress–energy tensor (source): $T_{\mu\nu} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{WCT}}}{\delta g^{\mu\nu}}$, with conservation $\nabla^{\mu} T_{\mu\nu} = 0$ from diffeomorphism invariance of S_{WCT} .
- Locked rest scale (local): $k_{\text{eff}}(x) := \langle \sigma \rangle_w$; inertial mass density $m_{\text{dens}} := \frac{\hbar}{c} k_{\text{eff}} \rho$; energy density $u := T_{00}$ in the local rest frame.

Q.2 Newtonian sector from the WCT source

Definition Q.1 (Gravitational potential and Lagrangian). Introduce a scalar potential $\phi(\mathbf{x}, t)$ with Lagrangian density

$$\mathcal{L}_\phi = +\frac{c^2}{8\pi G} (\nabla\phi)^2 + \frac{\phi}{c^2} T_{00}.$$

Proposition 17 (Poisson equation from variation). *Varying ϕ gives*

$$\nabla^2\phi = \frac{4\pi G}{c^2} T_{00}.$$

Proof. $\partial\mathcal{L}_\phi/\partial\phi = T_{00}/c^2$, $\partial\mathcal{L}_\phi/\partial(\partial_i\phi) = \frac{c^2}{4\pi G} \partial_i\phi$, hence Euler–Lagrange yields the stated Poisson equation. $\square$

Corollary 12 (Equivalence principle and redshift). *For any locked object with $E_{\text{rest}}^0 = mc^2$, set $E_{\text{rest}}(\mathbf{x}) = E_{\text{rest}}^0(1 + \phi/c^2)$. Then $\mathbf{F} = -\nabla E_{\text{rest}} = -m\nabla\phi$ and $\Delta f/f = \Delta\phi/c^2$. Thus free-fall acceleration $\mathbf{a} = -\nabla\phi$ is composition-independent.*

Q.3 Linearized spin-2 from WCT: universal coupling and $\gamma = 1$

Definition Q.2 (Metric perturbation and gauge). Let $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$, trace-reverse $\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, $h := \eta^{\alpha\beta}h_{\alpha\beta}$. Impose harmonic gauge $\partial^\mu h_{\mu\nu} = 0$.

Proposition 18 (Linearized field equations with WCT source). *The most general Lorentz-covariant, ghost-free quadratic tensor action that couples universally to $T^{\mu\nu}$ is the Fierz–Pauli form*

$$S_g^{(2)}[h] + S_{\text{int}}[h, \psi] = \frac{c^3}{64\pi G} \int d^4x \left(\partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} - 2\partial_\mu h^{\mu\lambda} \partial^\nu h_{\nu\lambda} + 2\partial_\mu h^{\mu\nu} \partial_\nu h - \partial_\lambda h \partial^\lambda h \right) + \frac{1}{2} \int d^4x h_{\mu\nu} T^{\mu\nu}.$$

Variation gives

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \quad \square := -c^{-2} \partial_t^2 + \nabla^2.$$

Corollary 13 (Newtonian limit and $\gamma = 1$). *For static weak fields with $T_{00} \gg T_{ij}$, choose $h_{00} = -2\phi/c^2$, $h_{ij} = -2\phi/c^2 \delta_{ij}$. Then $\bar{h}_{00} = -4\phi/c^2$, so $\nabla^2 \bar{h}_{00} = -16\pi G T_{00}/c^4$ reproduces Prop. 17. Equality of temporal/spatial coefficients implies $\gamma = 1$ (correct light bending/Shapiro delay).*

Q.4 From linear spin-2 to full GR by consistency

Theorem Q.3 (Self-coupling closure to Einstein–Hilbert). *Let $h_{\mu\nu}$ be the massless spin-2 of Prop. 18 with universal $h_{\mu\nu} T^{\mu\nu}$ coupling and $\nabla^\mu T_{\mu\nu} = 0$ from S_{WCT} . Demanding:*

- (a) *gauge consistency (nonlinear completion of $\partial^\mu \bar{h}_{\mu\nu} = 0$),*
- (b) *absence of ghosts and superluminal modes,*
- (c) *conservation of the total stress (matter + graviton),*

uniquely promotes $S_g^{(2)}$ to the Einstein–Hilbert action

$$S_g[g] = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g} R,$$

with field equations $G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$. Its linearization reproduces Prop. 18, and its static limit reproduces Prop. 17.

Proof sketch. Iterate the coupling to the stress of $h_{\mu\nu}$ itself (self-coupling) and enforce gauge symmetry order by order; the unique nonlinear completion with two derivatives and universal coupling is the Einstein–Hilbert action. The Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ ensures total stress conservation. $\square$

Q.5 Determination of G from WCT (long-wavelength Kubo)

Definition Q.4 (TT correlator and Kubo map). Let $\Pi^{\text{TT}}(\omega, \mathbf{k})$ be the retarded *transverse-traceless* (TT) correlator of T_{ij} in the locked background:

$$\Pi^{\text{TT}}(\omega, \mathbf{k}) := -i \int d^4x e^{i(\omega t - \mathbf{k} \cdot \mathbf{x})} \theta(t) \langle [T_{ij}^{\text{TT}}(x), T_{ij}^{\text{TT}}(0)] \rangle.$$

Proposition 19 (Low-energy matching). *If $\Pi^{\text{TT}}(\omega, \mathbf{k}) \sim \kappa_2 \mathbf{k}^2 + \mathcal{O}(\mathbf{k}^4, \omega^2)$ as $(\omega, \mathbf{k}) \rightarrow 0$, then matching the quadratic actions yields*

$$\frac{1}{16\pi G} = \frac{\kappa_2}{c^3}.$$

Thus G is fixed by the WCT long-wavelength TT stiffness κ_2 of the locked background.

Q.6 Worked example (single ring source)

Example Q.5 (Poisson from a thin ring). Take a thin ring of radius R in the xy -plane with line energy λ (J/m), so $T_{00}(\mathbf{x}) = \lambda \int_0^{2\pi} R d\theta \delta^{(3)}(\mathbf{x} - \mathbf{x}(\theta))$, $\mathbf{x}(\theta) = (R \cos \theta, R \sin \theta, 0)$. By Prop. 17, $\nabla^2 \phi = (4\pi G/c^2) T_{00}$, whose solution is

$$\phi(\mathbf{r}) = \frac{2G\lambda}{c^2} K\left(\frac{4Rr_\perp}{(R+r_\perp)^2 + z^2}\right) \frac{1}{\sqrt{(R+r_\perp)^2 + z^2}},$$

with $r_\perp = \sqrt{x^2 + y^2}$ and K the complete elliptic integral (standard ring potential). Near the axis ($r_\perp \ll R$), $\phi(z) \approx -\frac{2G\lambda}{c^2} \frac{1}{\sqrt{R^2 + z^2}}$, giving $a_z = -\partial_z \phi$ and redshift $\Delta f/f = \Delta \phi/c^2$.

Summary of closure. (i) $S_{\text{WCT}} \Rightarrow$ conserved $T_{\mu\nu}$; (ii) $\mathcal{L}_\phi \Rightarrow$ Poisson with WCT source (Newton + EP + redshift); (iii) universal spin-2 coupling $\Rightarrow$ Fierz–Pauli $\Rightarrow$ Einstein–Hilbert; (iv) G fixed by the TT stiffness κ_2 (Kubo map). This completes the bridge from curvature locking to macro (Newton/PPN/GW) and provides a concrete EFT route at quantum scales.

R Static TT Matching and Correct Normalization (G7)

Definition R.1 (TT stiffness κ_2). Let h_{ij}^{TT} be the transverse–traceless spatial metric perturbation and T_{ij}^{TT} the TT part of the stress. Define the static TT kernel of the medium by

$$T_{ij}^{\text{TT}}(\mathbf{k}) = \Pi^{\text{TT}}(0, \mathbf{k}) h_{ij}^{\text{TT}}(\mathbf{k}), \quad \Pi^{\text{TT}}(0, \mathbf{k}) = \kappa_2 k^2 + \mathcal{O}(k^4),$$

so that in real space $E_{\text{mat}}^{\text{TT}} = \frac{\kappa_2}{2} \int (\partial_\ell h_{ij}^{\text{TT}})^2 d^3x + \mathcal{O}(\nabla^4)$.

Proposition 20 (Kernel match $\Rightarrow 1/(16\pi G) = \kappa_2/c^4$). *In harmonic gauge the linearized Einstein equations read $\square \bar{h}_{\mu\nu} = -(16\pi G/c^4) T_{\mu\nu}$. In the static TT sector (trace-free so $\bar{h}^{\text{TT}} = h^{\text{TT}}$):*

$$k^2 h_{ij}^{\text{TT}} = \frac{16\pi G}{c^4} T_{ij}^{\text{TT}} \quad \implies \quad h_{ij}^{\text{TT}} = \frac{16\pi G}{c^4} \frac{1}{k^2} T_{ij}^{\text{TT}}.$$

Integrating out h gives the source–source energy $E_{\text{GR}}^{\text{TT}} = -\frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} T^{\text{TT}} \left(\frac{16\pi G}{c^4} \frac{1}{k^2} \right) T^{\text{TT}}$. From Definition R.1, the medium alone yields $E_{\text{mat}}^{\text{TT}} = -\frac{1}{2} \int \frac{d^3 k}{(2\pi)^3} T^{\text{TT}} \left(\frac{1}{\kappa_2} \frac{1}{k^2} \right) T^{\text{TT}}$. Equating kernels implies

$$\boxed{\frac{1}{16\pi G} = \frac{\kappa_2}{c^4}}.$$

Remark 5 (Replacement note). Wherever $\frac{1}{16\pi G} = \frac{\kappa_2}{c^4}$ appeared, replace by $\frac{1}{16\pi G} = \frac{\kappa_2}{c^4}$. All stiffness-based force formulas gain one factor of c .

S Force Laws: Equivalent Forms (G8)

Definition S.1 (Locked mass/energy). For ring i : radius $R_i > 0$, holonomy $\theta_i = \ell_i + \Phi_i/\Phi_0 + s_i \Gamma_{s,i}$, locked wavenumber $k_{\text{eff},i} = \theta_i/R_i$,

$$E_i = \hbar c k_{\text{eff},i} = \frac{\hbar c}{R_i} \theta_i, \quad m_i = \frac{E_i}{c^2} = \frac{\hbar}{c R_i} \theta_i.$$

Proposition 21 (Vector law and substitutions). Let $\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2$, $r = \|\mathbf{r}_{12}\|$, $\hat{\mathbf{r}}_{12} = \mathbf{r}_{12}/r$. Then

$$\boxed{\mathbf{F}_{12} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}_{12}}$$

and the following forms are equivalent:

$$\boxed{\mathbf{F}_{12} = -\hat{\mathbf{r}}_{12} \frac{G}{c^4} \frac{E_1 E_2}{r^2} = -\hat{\mathbf{r}}_{12} \frac{\hbar^2 c^2}{16\pi \kappa_2} \frac{\theta_1 \theta_2}{R_1 R_2} \frac{1}{r^2}}.$$

T Units Audit (G9)

Lemma T.1 (SI consistency). $[G] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, $[c] = \text{m s}^{-1}$, so $[c^4/G] = \text{kg m s}^{-2} = \text{N}$. Hence $[\kappa_2] = \text{N}$ and the identity $\frac{1}{16\pi G} = \frac{\kappa_2}{c^4}$ is dimensionally exact. Moreover, $F = \frac{1}{16\pi \kappa_2} \frac{E_1 E_2}{r^2}$ has units of newtons.

U Post-Newtonian Check (G10)

Proposition 22 ($\gamma = \beta = 1$). With universal coupling $\frac{1}{2} \int h_{\mu\nu} T^{\mu\nu}$ and the nonlinear self-coupling that uniquely promotes the quadratic spin-2 action to Einstein–Hilbert, the 1PN metric reads $g_{tt} = -(1 + 2\phi/c^2 + 2\phi^2/c^4) + \dots$, $g_{ij} = (1 - 2\phi/c^2)\delta_{ij} + \dots$, yielding PPN parameters $\gamma = 1$, $\beta = 1$.

V Observable Rigidity (G11)

Corollary 14 (Vacuum TT stiffness). Using $G_{\text{N}} = 6.67430 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$ and $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$,

$$\boxed{\kappa_2 = \frac{c^4}{16\pi G_{\text{N}}} \approx 2.41 \times 10^{42} \text{ N}}.$$

For rings with $E_i = \hbar c \theta_i/R_i$ the predicted Newton force is $F(r) = \frac{1}{16\pi \kappa_2} \frac{E_1 E_2}{r^2}$.

W Minimal Two-Body Extractors (G12)

Proposition 23 (Solve for G or κ_2 from $F(r)$). *Given a measured magnitude $F = \|\mathbf{F}_{12}\|$ at separation r ,*

$$\boxed{G = \frac{F r^2}{m_1 m_2} = \frac{F r^2 c^4}{E_1 E_2} = \frac{F r^2 c^2 R_1 R_2}{\hbar^2 \theta_1 \theta_2}}, \quad \boxed{\kappa_2 = \frac{1}{16\pi} \frac{E_1 E_2}{F r^2} = \frac{\hbar^2 c^2}{16\pi} \frac{\theta_1 \theta_2}{R_1 R_2} \frac{1}{F r^2}}.$$

X Computational Review Note

Several numerical and logical assessments in this paper were assisted by ChatGPT, an AI language model developed by OpenAI. The model was used to check order-of-magnitude estimates, structural coherence, and consistency with existing scientific literature. All factual claims remain the responsibility of the author.